

Summer Bootcamp – Session 3

Hands-On Optimization Exercises

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Instructions

Exercises 1–3 concern one-dimensional unconstrained optimization. Exercises 4–6 move to several variables. Exercises 7–9 introduce equality and inequality constraints. Exercises 10–11 are algorithmic implementations. Exercises 12–14 apply the theory to portfolio choice. The integrated Python project is provided separately in the **Portfolio Optimization Project** folder.

For analytical exercises, justify every candidate and distinguish necessary from sufficient conditions. For Python work, use `NumPy`, `pandas`, `matplotlib`, and, where explicitly allowed, `scipy.optimize`. Set all random seeds and report convergence criteria. Do not use future test observations when estimating a portfolio.

Notation and supplemental definitions

The following notation completes the material explicitly introduced in the session notes.

- $\mathbf{1} = (1, \dots, 1)^\top \in \mathbb{R}^d$ is the vector of ones. For a symmetric positive-definite matrix Q , write $\lambda_{\min}(Q)$ and $\lambda_{\max}(Q)$ for its smallest and largest eigenvalues, and $\kappa(Q) = \lambda_{\max}(Q)/\lambda_{\min}(Q)$ for its spectral condition number.
- A matrix $A \in \mathbb{R}^{m \times d}$ has *full row rank* when $\text{rank}(A) = m$. Its singular values are the nonnegative square roots of the eigenvalues of $A^\top A$.
- For a descent direction p_k , Armijo backtracking starts from a trial step $\alpha > 0$ and repeatedly replaces α by $\rho\alpha$, with $0 < \rho < 1$, until

$$f(x_k + \alpha p_k) \leq f(x_k) + c\alpha \nabla f(x_k)^\top p_k, \quad 0 < c < 1.$$

- A centered finite-difference approximation of the j th gradient component is

$$\partial_j f(x) \approx \frac{f(x + he_j) - f(x - he_j)}{2h},$$

where e_j is the j th coordinate vector and $h > 0$ is small.

- A first-order perturbation keeps only terms linear in a small change. In particular,

$$(\Sigma + E)^{-1} \approx \Sigma^{-1} - \Sigma^{-1} E \Sigma^{-1}$$

when E is small and Σ is invertible.

- *In-sample* data are used to estimate or select a model; *out-of-sample* data are held back and used only for final evaluation. A hat, as in $\hat{\mu}$ or $\hat{\Sigma}$, denotes an estimate.

1 Part 1 – One-Dimensional Unconstrained Optimization

Exercise 1 – Critical points are only candidates. For each function below, find every critical point, classify it as a local minimum, local maximum, or neither, and determine whether a global optimizer exists on the stated

domain:

$$f_1(x) = x^4 - 4x^2 + 3, \quad x \in \mathbb{R},$$

$$f_2(x) = x^3 - 3x + 1, \quad x \in [-2, 2], \quad f_3(x) = \log(1 + e^x) - \frac{3}{4}x, \quad x \in \mathbb{R}.$$

1. Use first- and second-order conditions where they apply.
2. For f_2 , include the boundary points in the global comparison.
3. Prove that f_3 is strictly convex and compute its unique minimizer.
4. Sketch the three functions and mark all candidates.

Exercise 2 – A parameter changes the number of local optima. Let

$$f_a(x) = \frac{1}{4}x^4 - \frac{a}{2}x^2 + x, \quad a > 0.$$

1. Show that stationary points solve $x^3 - ax + 1 = 0$.
2. Study the derivative of this cubic and determine for which values of a the objective has one stationary point and for which values it has three.
3. In the three-point regime, classify the stationary points without solving the cubic explicitly.
4. Without computing decimal approximations, explain how the sign pattern of f'_a orders the local extrema in the three-point regime.
5. Explain why $f'_a(x) = 0$ alone cannot identify the global solution and give a rigorous comparison strategy.

Exercise 3 – Newton's method as a local model. Consider

$$f(x) = e^x - 3x + \frac{1}{4}x^2.$$

1. Prove that f has a unique global minimizer.
2. Write Newton's iteration for solving $f'(x) = 0$.
3. Express the Newton direction as the minimizer of the local quadratic model of f at x_k .
4. Determine conditions on x_k under which the full Newton step is a descent direction.
5. Explain how Armijo backtracking modifies the update and why it improves global robustness. Numerical trajectories belong to Exercise 10.

2 Part 2 – Multidimensional Unconstrained Optimization

Exercise 4 – Quadratic optimization and conditioning. Let

$$f(x) = \frac{1}{2}x^\top Qx - c^\top x, \quad Q = Q^\top \succ 0, \quad x, c \in \mathbb{R}^d.$$

1. Compute ∇f and $\nabla^2 f$.
2. Proceed in three steps:
 - (a) use $Q \succ 0$ to prove that f is strictly convex;
 - (b) write the stationarity equation as the linear system $Qx^* = c$;
 - (c) explain why $Q \succ 0$ implies that Q is invertible and conclude that this system has exactly one solution.The notation $x^* = Q^{-1}c$ may be used for the theoretical formula, but a numerical implementation should solve $Qx = c$ rather than form Q^{-1} .
3. Complete the square to express $f(x) - f(x^*)$ as a quadratic form.
4. In an orthonormal eigenbasis of Q , interpret the eigenvalues as principal curvatures.
5. Let $e_k = x_k - x^*$ and consider gradient descent with a constant step α . Detail the role of the condition number as follows:
 - (a) show in an orthonormal eigenbasis of Q that each error coordinate satisfies

$$(e_{k+1})_i = (1 - \alpha\lambda_i)(e_k)_i;$$

- (b) explain why a large ratio $\kappa(Q) = \lambda_{\max}(Q)/\lambda_{\min}(Q)$ produces elongated level sets and very different contraction factors across directions;
- (c) recall the stability condition $0 < \alpha < 2/\lambda_{\max}(Q)$ and compare qualitatively the cases $\kappa(Q) \approx 1$ and $\kappa(Q) \gg 1$.

Exercise 5 – Least squares, ridge regression, and conditioning. Let $A \in \mathbb{R}^{n \times d}$ and $b \in \mathbb{R}^n$.

1. Minimize $\frac{1}{2} \|Ax - b\|_2^2$ by deriving the normal equations.
2. Give a necessary and sufficient condition for uniqueness. Interpret the residual geometrically when the normal equations hold.
3. For $\lambda > 0$, solve

$$\min_x \frac{1}{2} \|Ax - b\|_2^2 + \frac{\lambda}{2} \|x\|_2^2.$$

4. Here, *rank deficient* means that A does not have full column rank:

$$\text{rank}(A) < d.$$

Equivalently, the columns of A are linearly dependent and $A^\top A$ is singular. Show that ridge regression nevertheless has a unique solution by proving that $A^\top A + \lambda I \succ 0$ for every $\lambda > 0$.

5. Use the singular values of A to explain how regularization changes the condition number and creates a bias–stability trade-off.

Exercise 6 – Second-order conditions in several dimensions. For each function, find all stationary points and classify them:

$$\begin{aligned} f_1(x, y) &= x^2 + xy + y^2 - 4x + 2y, \\ f_2(x, y) &= x^4 + y^4 - 2x^2 - 4y^2, \quad f_3(x, y) = x^2 - y^2 + 2xy. \end{aligned}$$

1. Compute each Hessian and use its eigenvalues or principal minors.
2. For degenerate candidates of f_2 , inspect the objective along useful directions when the Hessian test is inconclusive.
3. Identify which local minima are also global and justify the answer.
4. Plot contours for the three objectives on a domain of your choice.

3 Part 3 – Multidimensional Constrained Optimization

Exercise 7 – Projection onto an affine constraint. Let $z, a \in \mathbb{R}^d$, with $a \neq 0$, and $C = \{x \in \mathbb{R}^d : a^\top x = b\}$.

1. Formulate $\text{proj}_C(z)$ as a constrained quadratic problem.
2. Solve it with a Lagrange multiplier.
3. Obtain a closed formula for $\text{proj}_C(z)$ and verify that it returns z when $z \in C$.
4. Verify geometrically that $z - \text{proj}_C(z)$ is normal to C .
5. Extend the derivation to $C = \{x \in \mathbb{R}^d : Ax = b\}$, where $A \in \mathbb{R}^{m \times d}$. The matrix A has *full row rank* when

$$\text{rank}(A) = m,$$

meaning that its m rows are linearly independent. Explain why this implies that $AA^\top$ is invertible, then derive the projection formula.

Exercise 8 – Equality constraints, duality, and a saddle point. Consider the generic equality-constrained quadratic problem

$$\min_{x \in \mathbb{R}^d} \frac{1}{2} x^\top Q x + c^\top x \quad \text{subject to} \quad Ax = b,$$

where $Q \in \mathbb{R}^{d \times d}$ is symmetric positive definite and $A \in \mathbb{R}^{m \times d}$ has full row rank.

1. Write the Lagrangian and the KKT linear system.
2. Eliminate x and derive formulas for λ^* and x^* .

3. Compute the dual function

$$q(\lambda) = \inf_x \mathcal{L}(x, \lambda)$$

explicitly and show directly that it is concave.

4. Prove weak duality for every feasible x and every multiplier λ .
5. Verify strong duality and the saddle inequalities symbolically.
6. With the convention $\mathcal{L}(x, \lambda) = f(x) + \lambda^\top(Ax - b)$, show that the first-order sensitivity of the optimal value to b is $-\lambda^*$.

Exercise 9 – Projection onto a half-space with KKT. Let $z, a \in \mathbb{R}^d$, $a \neq 0$, and solve

$$\min_x \frac{1}{2} \|x - z\|_2^2 \quad \text{subject to} \quad a^\top x \leq b.$$

1. Draw the geometry in dimension two and identify the unconstrained minimizer.
2. Write every KKT condition, including dual feasibility and complementary slackness.
3. Treat separately the cases $a^\top z \leq b$ and $a^\top z > b$. Derive the optimizer and multiplier in each case.
4. Explain why the KKT conditions are sufficient here.
5. Show that the projection map can be written using the positive part $(u)_+ = \max(u, 0)$ and interpret the active-set change.

4 Part 4 – Python Algorithms

Exercise 10 – Implement gradient descent from scratch. Implement

$$x_{k+1} = x_k - \alpha_k \nabla f(x_k)$$

without calling an optimization routine. For the quadratic test, set a random seed and construct $Q = M^\top M + 0.1I$ and c with $d = 20$; retain the exact solution from Exercise 4 as a benchmark. Also test the Rosenbrock function

$$r(x, y) = 100(y - x^2)^2 + (1 - x)^2.$$

1. Write a reusable function accepting **f**, **grad**, an initial point, a step-size rule, a tolerance, and a maximum iteration count.
2. Return the entire history of iterates, objective values, gradient norms, and accepted step sizes.
3. For the quadratic, compare constant steps $1/\lambda_{\max}(Q)$, $2/(\lambda_{\min}(Q) + \lambda_{\max}(Q))$, and one unstable step satisfying $\alpha > 2/\lambda_{\max}(Q)$.
4. Add Armijo backtracking with $c = 10^{-4}$ and $\rho = 1/2$, then test the code on Rosenbrock from $(-1.2, 1)^\top$.
5. Plot objective gap and gradient norm on logarithmic axes. Plot the iterates over objective contours.
6. Check the analytical gradient with centered finite differences before running the optimizer.

Algorithmic deliverable

Submit the implementation, two convergence tables, and four diagnostic plots. State the stopping rule explicitly. A plot alone is not evidence of convergence: compare with the known solution whenever one is available.

Exercise 11 – Implement primal–dual gradient descent–ascent. Use the equality-constrained quadratic from Exercise 8. Generate a reproducible instance with $d = 20$, $m = 3$, $Q = M^\top M + 0.1I$, and a full-row-rank matrix A . Draw c and a reference point x^{ref} , then set $b = Ax^{\text{ref}}$ so feasibility is guaranteed. Its Lagrangian is

$$\mathcal{L}(x, \lambda) = \frac{1}{2} x^\top Q x + c^\top x + \lambda(Ax - b).$$

Implement the simultaneous iteration

$$x_{k+1} = x_k - \alpha(Qx_k + c + A^\top \lambda_k), \quad \lambda_{k+1} = \lambda_k + \beta(Ax_k - b).$$

1. Start from $x_0 = 0$, $\lambda_0 = 0$ and test several pairs (α, β) .
2. At every iteration record the primal error, feasibility, and stationarity residuals

$$\|x_k - x^*\|, \quad \|Ax_k - b\|, \quad \|Qx_k + c + A^\top \lambda_k\|,$$

as well as the Lagrangian value.

3. Find one stable and one oscillatory parameter choice. Plot the trajectories and explain the difference.
4. Implement the augmented Lagrangian

$$\mathcal{L}_\rho(x, \lambda) = \mathcal{L}(x, \lambda) + \frac{\rho}{2} \|Ax - b\|_2^2$$

and its descent-ascent updates. Study $\rho = 0.1, 1, 10$.

5. Compare all numerical solutions with the exact KKT solution from Exercise 8.
6. Optional: define $z = (x, \lambda)$ and the saddle operator $F(z) = (\nabla_x \mathcal{L}, -\nabla_\lambda \mathcal{L})$. Implement the extragradient update

$$\tilde{z}_k = z_k - \eta F(z_k), \quad z_{k+1} = z_k - \eta F(\tilde{z}_k),$$

then compare its stability with simultaneous descent-ascent.

5 Part 5 – Portfolio Optimization

Let $R \in \mathbb{R}^d$ be the random vector of annual asset returns, with mean $\mu = \mathbb{E}[R]$ and covariance matrix $\Sigma = \text{Cov}(R)$. Unless otherwise stated, portfolio weights satisfy $\mathbf{1}^\top w = 1$. For risk-free rate r_f , define

$$R(w) = \mu^\top w, \quad \sigma(w) = \sqrt{w^\top \Sigma w}, \quad \text{SR}(w) = \frac{R(w) - r_f}{\sigma(w)}.$$

Exercise 12 – Risk, return, and the Sharpe ratio. Let $\mu \in \mathbb{R}^d$, $\Sigma \succ 0$, and $r_f \in \mathbb{R}$.

1. Prove the formulas for portfolio return and variance from the random return vector R .
2. Derive the global minimum-variance portfolio allowing short sales:

$$w_{\text{GMV}} = \frac{\Sigma^{-1} \mathbf{1}}{\mathbf{1}^\top \Sigma^{-1} \mathbf{1}}.$$

3. Show that the tangency direction is proportional to $\Sigma^{-1}(\mu - r_f \mathbf{1})$ and explain how full investment fixes its normalization when the denominator is nonzero.
4. Compare the objectives answered by equal weight, GMV, and tangency portfolios. State when each is a sensible benchmark.
5. Explain why long-only constraints can invalidate the closed formulas and identify the KKT conditions required instead.

Exercise 13 – Derive the efficient frontier. Consider

$$\min_w \frac{1}{2} w^\top \Sigma w \quad \text{subject to} \quad \mathbf{1}^\top w = 1, \quad \mu^\top w = m$$

for a feasible target return m .

1. Derive the KKT system for a fixed target m .
2. With

$$C = \begin{pmatrix} \mathbf{1}^\top \\ \mu^\top \end{pmatrix}, \quad d_m = \begin{pmatrix} 1 \\ m \end{pmatrix},$$

derive $w(m) = \Sigma^{-1} C^\top (C \Sigma^{-1} C^\top)^{-1} d_m$.

3. Show that the optimal variance is a quadratic function of m and deduce the shape of the frontier in mean-standard-deviation space.
4. Explain qualitatively why long-only constraints and position caps create kinks and active-set changes.
5. State the feasibility and positive-semidefiniteness checks a numerical solver should perform before tracing a frontier.

Exercise 14 – Estimation error and robust weights. Suppose an estimate $(\hat{\mu}, \hat{\Sigma})$ replaces the population parameters, with $\hat{\Sigma} \succ 0$. Define the unconstrained tangency direction

$$q(\hat{\mu}, \hat{\Sigma}) = \hat{\Sigma}^{-1}(\hat{\mu} - r_f \mathbf{1}).$$

1. Study the sensitivity of q to mean-estimation error. Throughout this question, keep $\hat{\Sigma}$ and r_f fixed. Let $h \in \mathbb{R}^n$ denote an estimation-error direction and replace $\hat{\mu}$ by $\hat{\mu} + \varepsilon h$.
 - (a) Write $q(\hat{\mu} + \varepsilon h, \hat{\Sigma})$ and isolate the first-order change $q(\hat{\mu} + \varepsilon h, \hat{\Sigma}) - q(\hat{\mu}, \hat{\Sigma})$.
 - (b) Deduce the directional derivative $D_{\mu} q(\hat{\mu}, \hat{\Sigma})[h]$ and the Jacobian of q with respect to $\hat{\mu}$.
 - (c) Use an eigendecomposition $\hat{\Sigma} = U \text{diag}(\lambda_1, \dots, \lambda_n) U^T$. Decompose h in the orthonormal eigenbasis and determine how the component along an eigenvector associated with λ_i is transformed.
 - (d) Give an upper bound for the relative or absolute change in q in terms of $\lambda_{\min}(\hat{\Sigma})$ and $\|h\|_2$. Explain why a mean-estimation error aligned with a small-eigenvalue direction can produce a large change in the portfolio direction.
 - (e) Optional: if the fully invested weights are $w = q/(\mathbf{1}^T q)$, explain why the normalization may create an additional instability when $\mathbf{1}^T q$ is close to zero.
2. Explain why GMV weights are insensitive to mean estimation but remain sensitive to covariance estimation.
3. Define the ridge-regularized covariance

$$\hat{\Sigma}_{\delta} = \hat{\Sigma} + \delta I, \quad \delta \geq 0.$$

Show how its eigenvalues and condition number depend on δ .

4. Compare in-sample predicted volatility with out-of-sample realized volatility. Explain why optimizing estimated expected returns is particularly unstable.
5. Propose a transparent validation rule for selecting δ without looking at the final test sample. Reserve all numerical comparisons for the separate portfolio project.

Separate Portfolio Optimization Project

The integrated project is in the separate Portfolio Optimization Project folder. Use its detailed statement and `portfolio_project_skeleton.py`, which contains the staged questions, validation checks, expected outputs, and explicit TODO blocks.